Proxmox VE + Ceph + ZFS Cluster — 57-Step Live Test Report

CISAI-PVE · PROX22 / PROX23 / PROX24

Generated: 2026-09-19 15:11:08 | **Source:** fresh_live_test_results.json | **Cluster:** 3-node, 12 OSDs, Ceph Reef, ZFS RAIDZ2 (zfs-ha) | **Method:** Live fault injection with auto-recovery and HEALTH_OK verification

57	0	6	57
PASSED	FAILED	SKIPPED	TOTAL

Fault-Injection & Resilience Test Matrix (57 steps)

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT

1	**[Step 1.0] Remove 1 Disk from 1 Server**
	Remove 1 disk (osd.0) from prox22, verify cluster still serves data.
		`osd.0 is already down.`
		`marked out osd.0.`
		`marked in osd.0.`
			`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_WARN Degraded data redundancy: 340/3126 objects degraded (10.877%), 4 pgs degraded services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 11 up (since 5s), 11 in (since 4s); 1 remapped pgs data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 200 TiB / 200 TiB avail pgs: 340/3126 objects degraded (10.877%) 28 active+clean 2 active+recovery_wait+undersized+degraded+remapped 2 active+recovering+undersized+degraded+remapped 1 stale+active+clean io: recovery: 51 MiB/s, 124 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: -c: line 4: syntax error near unexpected token `(' bash: -c: line 4: ` Degraded data redundancy:	5s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			340/3126 objects degraded (10.877%), 4 pgs degraded' `HEALTH_WARN Degraded data redundancy: 340/3126 objects degraded (10.877%), 4 pgs degraded [WRN] PG_DEGRADED: Degraded data redundancy: 340/3126 objects degraded (10.877%), 4 pgs degraded pg 3.0 is active+recovery_wait+undersized+degraded+remapped, acting [5,4] pg 4.1 is active+recovery_wait+undersized+degraded+remapped, acting [2,1] pg 4.9 is active+recovering+undersized+degraded+remapped, acting [5,1] pg 4.1e is active+recovering+undersized+degraded+remapped, acting [7,5]` → bash: -c: line 1: syntax error near unexpected token `(' bash: -c: line 1: `HEALTH_WARN Degraded data redundancy: 340/3126 objects degraded (10.877%), 4 pgs degraded' `ID CLASS WEIGHT TYPE NAME STATUS REWEIGHT PRI-AFF -1 218.28479 root default -3 72.76160 host prox22 0 hdd 18.19040 osd.0 down 0 1.00000 3 hdd 18.19040 osd.3 up 1.00000 1.00000 6 hdd 18.19040 osd.6 up 1.00000 1.00000 9 hdd 18.19040 osd.9 up 1.00000 1.00000 -5 72.76160 host prox23 1 hdd 18.19040 osd.1 up 1.00000 1.00000 4 hdd 18.19040 osd.4 up 1.00000 1.00000 7 hdd 18.19040 osd.7 up 1.00000 1.00000 10 hdd 18.19040 osd.10 up 1.00000 1.00000 -7 72.76160 host prox24 2 hdd 18.19040 osd.2 up 1.00000 1.00000 5 hdd 18.19040 osd.5 up 1.00000 1.00000 8 hdd 18.19040 osd.8 up 1.00000 1.00000 11 hdd 18.19040 osd.11 up 1.00000 1.00000` → bash: line 1: ID: command not found bash: line 2: -1:			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			command not found bash: line 3: -3: command not found bash: line 4: 0: command not found bash: line 5: 3: command not found bash: line 6: 6: command not found bash: line 7: 9: command not found bash: line 8: -5: command not found bash: line 9: 1:… `final_health` → HEALTH_OK			
2	**[Step 2.0] Remove 1 Disk from 1 Server (verify availability)** While osd.0 is down, verify data availability on remaining OSDs.	`28` `12 osds: 12 up (since 2s), 12 in (since 6s); epoch: e1784`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22 (active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 2s), 12 in (since 6s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 13 GiB used, 218 TiB / 218 TiB avail pgs: 15.152% pgs not active 28 active+clean 5 remapped+peering io: client: 121 KiB/s wr, 0 op/s rd, 5 op/s wr recovery: 52 MiB/s, 1 keys/s, 154 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK	10s	HEALTH_OK	PASS
3	**[Step 3.0] Remove 1 Disk from 1 Server (self-heal check)** Verify cluster begins self-healing after single OSD loss.	`health: HEALTH_OK pools: 2 pools, 33 pgs pgs: 33 active+clean`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22 (active, since 2`	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 14s), 12 in (since 18s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 192 B/s rd, 3 KiB/s wr, 0 op/s rd, 0 op/s wr recovery: 10 MiB/s, 29 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK			
4	**[Step 4.0] Remove 2 Disks from 1 Server** Remove 2 disks (osd.0,3) from prox22 simultaneously.	`osd.0 is already down. osd.3 is already down.` `marked out osd.0. marked out osd.3.` `marked in osd.0. marked in osd.3.`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_WARN Reduced data availability: 2 pgs inactive, 2 pgs peering services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 10 up (since 6s), 10 in (since 5s); 10 remapped pgs data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 11 GiB used, 182 TiB / 182 TiB avail pgs: 33.333% pgs not active 21 active+clean 11 peering 1 stale+active+clean io: client: 441 B/s rd, 0 op/s rd, 0 op/s wr recovery: 82 MiB/s, 223 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found	15s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			bash: line 4: Reduced: command not found bash: line 6: services:: command not found bash: -c: line 7: syntax error near unexpected token `(' bash: -c: line 7: ` mon: 3 daemons, … `HEALTH_WARN Reduced data availability: 2 pgs inactive, 2 pgs peering [WRN] PG_AVAILABILITY: Reduced data availability: 2 pgs inactive, 2 pgs peering pg 4.0 is stuck peering for 95s, current state peering, last acting [10,6,8] pg 4.11 is stuck peering for 93s, current state peering, last acting [7,2,6]` → bash: line 1: HEALTH_WARN: command not found bash: line 2: [WRN]: command not found bash: line 3: pg: command not found bash: line 4: pg: command not found `ID CLASS WEIGHT TYPE NAME STATUS REWEIGHT PRI-AFF -1 218.28479 root default -3 72.76160 host prox22 0 hdd 18.19040 osd.0 down 0 1.00000 3 hdd 18.19040 osd.3 down 0 1.00000 6 hdd 18.19040 osd.6 up 1.00000 1.00000 9 hdd 18.19040 osd.9 up 1.00000 1.00000 -5 72.76160 host prox231 hdd 18.19040 osd.1 up 1.00000 1.00000 4 hdd 18.19040 osd.4 up 1.00000 1.00000 7 hdd 18.19040 osd.7 up 1.00000 1.00000 10 hdd 18.19040 osd.10 up 1.00000 1.00000 -7 72.76160 host prox24 2 hdd 18.19040 osd.2 up 1.00000 1.00000 5 hdd 18.19040 osd.5 up 1.00000 1.00000 8 hdd 18.19040 osd.8 up 1.00000 1.00000 11 hdd 18.19040 osd.11 up 1.00000 1.00000` → bash: line 1: ID: command not found bash: line 2: -1: command not found bash: line 3: -3: command not found bash: line 4: 0: command not found			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			bash: line 5: 3: command not found bash: line 6: 6: command not found bash: line 7: 9: command not found bash: line 8: -5: command not found bash: line 9: 1:… `final_health` → HEALTH_OK			
5	**[Step 5.0] Remove 2 Disks from 1 Server (verify availability)** Verify data availability with 2 OSDs down.	`33 12 osds: 12 up (since 11s), 12 in (since 16s); epoch: e1797`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 12s), 12 in (since 17s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 13 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 204 B/s rd, 3.3 KiB/s wr, 0 op/s rd, 0 op/s wr` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
6	**[Step 6.0] Remove 2 Disks from 1 Server (self-heal check)** Verify self-healing with 2 OSDs down.	`health: HEALTH_OK pools: 2 pools, 33 pgs pgs: 33 active+clean`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 13s), 12 in (since 18s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usa`	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`ge: 13 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 204 B/s rd, 3.3 KiB/s wr, 0 op/s rd, 0 op/s wr` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK			

7	**[Step 7.0] Remove 3 Disks from 1 Server**
	Remove 3 disks (osd.0,3,6) from prox22.
		`osd.0 is already down. osd.3 is already down. osd.6 is already down.`
		`marked out osd.0. marked out osd.3. marked out osd.6.`
		`marked in osd.0. marked in osd.3. marked in osd.6.`
			`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_WARN Degraded data redundancy: 4542/3126 objects degraded (145.298%), 20 pgs degraded services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 9 up (since 6s), 9 in (since 5s); 16 remapped pgs data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 9.9 GiB used, 164 TiB / 164 TiB avail pgs: 4542/3126 objects degraded (145.298%) 15 active+recovery_wait+undersized+degraded+remapped 13 active+clean 3 active+undersized+degraded+remapped+backfill_wait 1 active+undersized+degraded 1 active+recovering+undersized+degraded+remapped io: client: 419 B/s rd, 0 op/s rd, 0 op/s wr recovery: 28 MiB/s, 30 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found	15s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			bash: line 3: health:: command not found bash: -c: line 4: syntax error near unexpected token `(' bash: -c: line 4: ` Degraded data redundancy: 4542/3126 objects degraded (145.298%), 20 pgs degraded' HEALTH_WARN Degraded data redundancy: 4542/3126 objects degraded (145.298%), 20 pgs degraded [WRN] PG_DEGRADED: Degraded data redundancy: 4542/3126 objects degraded (145.298%), 20 pgs degraded pg 3.0 is active+recovery_wait+undersized+degraded+remapped, acting [5,4] pg 4.0 is active+recovery_wait+undersized+degraded+remapped, acting [10,8] pg 4.1 is active+recovery_wait+undersized+degraded+remapped, acting [2,1] pg 4.2 is active+recovery_wait+undersized+degraded+remapped, acting [8,10] pg 4.3 is active+undersized+degraded+remapped+backfill_wait, acting [11,7] pg 4.4 is active+recovery_wait+undersized+degraded+remapped, acting [8,1] pg 4.7 is active+undersized+degraded+remapped+backfill_wait, acting [5,7] pg 4.9 is active+recovery_wait+undersized+degraded+remapped, acting [5,1] pg 4.a is active+recovery_wait+undersized+degraded+remapped, acting [1,2] pg 4.b is active+recovery_wait+undersized+degraded+remapped, acting [1,11] pg 4.d is active+recovery_wait+undersized+degraded+remapped, acting [4,2] pg 4.11 is active+recovery_wait+undersized+degraded+remapped, acting [7,2] pg 4.15 is active+recovery_wait+undersized+degraded+remapped, acting [5,1] pg 4.16 is			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			active+recovering+undersized+degraded+remapped, acting [10,11] pg 4.19 is active+recovery_wait+undersized+degraded+remapped, acting [2,1] pg 4.1a is active+undersized+degraded+remapped+backfill_wait, acting [4,5] pg 4.1b is active+recovery_wait+undersized+degraded+remapped, acting [4,5] pg 4.1c is active+recovery_wait+undersized+degraded+remapped, acting [2,4] pg 4.1d is active+undersized+degraded, acting [2,1] pg 4.1e is active+recovery_wait+undersized+degraded+remapped, acting [7,5] → bash: -c: line 1: syntax error near unexpected token `(' bash: -c: line 1: `HEALTH_WARN Degraded data redundancy: 4542/3126 objects degraded (145.298%), 20 pgs degraded' ID CLASS WEIGHT TYPE NAME STATUS REWEIGHT PRI-AFF -1 218.28479 root default -3 72.76160 host prox22 0 hdd 18.19040 osd.0 down 0 1.00000 3 hdd 18.19040 osd.3 down 0 1.00000 6 hdd 18.19040 osd.6 down 0 1.00000 9 hdd 18.19040 osd.9 up 1.00000 1.00000 -5 72.76160 host prox23 1 hdd 18.19040 osd.1 up 1.00000 1.00000 4 hdd 18.19040 osd.4 up 1.00000 1.00000 7 hdd 18.19040 osd.7 up 1.00000 1.00000 10 hdd 18.19040 osd.10 up 1.00000 1.00000 -7 72.76160 host prox24 2 hdd 18.19040 osd.2 up 1.00000 1.00000 5 hdd 18.19040 osd.5 up 1.00000 1.00000 8 hdd 18.19040 osd.8 up 1.00000 1.00000 11 hdd 18.19040 osd.11 up 1.00000 1.00000 → bash: line 1: ID: command not found bash: line 2: -1: command not found bash: line 3: -3:			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			command not found bash: line 4: 0: command not found bash: line 5: 3: command not found bash: line 6: 6: command not found bash: line 7: 9: command not found bash: line 8: -5: command not found bash: line 9: 1:… `final_health` → HEALTH_OK			
8	**[Step 8.0] Remove 3 Disks from 1 Server (verify availability)** Verify data availability with 3 OSDs down.	`33 12 osds: 12 up (since 8s), 12 in (since 16s); epoch: e1813`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 8s), 12 in (since 17s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 301 B/s rd, 86 KiB/s wr, 0 op/s rd, 3 op/s wr recovery: 0 B/s, 0 keys/s, 0 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
9	**[Step 9.0] Remove 3 Disks from 1 Server (self-heal check)** Verify self-healing with 3 OSDs down.	`health: HEALTH_OK pools: 2 pools, 33 pgs pgs: 33 active+clean`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 10s), 1`	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`2 in (since 18s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 255 B/s rd, 76 KiB/s wr, 0 op/s rd, 4 op/s wr recovery: 0 B/s, 0 keys/s, 0 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK			
10	**[Step 10.0] Slot one to Slot Three (physical slot identity)** Verify physical drive slot identity across 3 nodes (read-only).	`NAME UUID SERIAL WWN SIZE sda 7TRgU0-ywAB-QoC8-FZPd-MZWU-Ww7D-j50HS1 ZYDGFEW6 0x5000c500f588273b 18.2T └─ceph--d1246597--45f6--40a6--ad11--fb6c122df899-osd--block--d1dcb9a5--e8bf--4d3c--8d7d--6a7ea9446302 18.2T sdb HlHJmS-Wc0q-6uXk-CJmu-8k2d-btTe-nkHx4d ZYDG4WXP 0x5000c500f588885ab 18.2T └─ceph--3ec8f2b7--632e--4c25--9fd2--bd16d3484e93-osd--block--edad6ac4--f97b--4d61--b64b--b09a37d57062 18.2T sdc KBMC8n-X0vu-8g5g-Clyz-WfCh-zxvW-RRN73Z ZYDG0CL3 0x5000c500f5880ffb 18.2T └─ceph--e1ef2e00--3344--40d5--ba42--79cb70faab2e-osd--block--67725c87--56ed--42a5--8511--19d7371fe401 18.2T sdd EQu8lB-prDB-S6ic-lpoq-hOEt-exHj-Pi4c5C ZYDFZ35V 0x5000c500f58810f3 18.2T └─ceph--fdcc0b24--192f--4148--97e5--3af6666d8ac2-osd--block--f28de884--5aae--4e0f--af41--6ac0634fba06 18.2T sde ZYDG62WJ 0x5000c500f5881f8b 18.2T ├─sde1 92800206 67378897418 0x5000c500f5881f8b 18.2T └─sde9 0x5000c500f5881f8b 8M sdf ZYDG2VVR 0x5000c500f589219b 1`	`sda ZYDGFEW6 0x5000c500f588273b sdb ZYDG4WXP 0x5000c500f58885ab sdc ZYDG0CL3 0x5000c500f5880ffb sdd ZYDFZ35V 0x5000c500f58810f3 sde ZYDG62WJ 0x5000c500f5881f8b ├─sde1 0x5000c500f5881f8b └─sde9 0x5000c500f5881f8b sdf ZYDG2VVR 0x5000c500f589219b ├─sdf1 0x5000c500f589219b └─sdf9 0x5000c500f589219b sdg ZYDFV3PC 0x5000c500f587c8af ├─sdg1 0x5000c500f587c8af └─sdg9 0x5000c500f587c8af sdh ZYDGDMT3 0x5000c500f5889713 ├─sdh1 0x5000c500f5889713 └─sdh9 0x5000c500f5889713 sdi ZYDGBV46 0x5000c500f588891f ├─sdi1 0x5000c500f588891f └─sdi9 0x5000c500f588891f sdj ZYDG6AWH 0x5000c500f5893a8b` → bash: line 1: sda: command not found bash: line 2: sdb: command not found bash: line 3: sdc: command not found bash: line 4: sdd: command not found bash: line 5: sde: command not found	—	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
		8.2T ├─sdf1 92800206 67378897418 0x5000c5 00f589219b 18.2T └─s df9 0x5000c500f58921 9b 8M sdg ZYDFV3PC 0 x5000c500f587c8af 1 8.2T ├─sdg1 92800206 67378897418 0x5000c5 00f587c8af 18.2T └─s dg9 0x5000c500f587c8 af 8M sdh ZYDGDMT3 0 x5000c500f5889713 1 8.2T ├─sdh1 92800206 67378897418 0x5000c5 00f5889713 18.2T └─s dh9 0x5000c500f58897 13 8M sdi ZYDGBV46 0 x5000c500f588891f 1 8.2T ├─sdi1 92800206 67378897418 0x5000c5 00f588891f 18.2T └─s di9 0x5000c500f58889 1f 8M sdj ZYDG6AWH 0 x5000c500f5893a8b 1 8.2T ├─sdj1 92800206 67378897418 0x5000c5 00f5893a8b 18.2T └─s dj9 0x5000c500f5893a 8b 8M sdk ZYDFWL9E 0 x5000c500f58801ff 1 8.2T ├─sdk1 92800206 67378897418 0x5000c5 00f58801ff 18.2T └─s dk9 0x5000c500f58801 ff 8M sdl ZYDFW4AP 0 x5000c500f588810b 1 8.2T sdm 00153b996d2 78a193200adbc58809b4 e 0x6b4e9b8058bcad00 32198a276d993b15 89 3.8G ├─sdm1 0x6b4e9b 8058bcad0032198a276d 993b15 1007K ├─sdm2 FD0A-D141 0x6b4e9b80 58bcad0032198a276d99 3b15 1G └─sdm3 CkWqE B-NXAO-DeXJ-0jNz-DaL 0-fgGa-Urtpy1 0x6b4e 9b8058bcad0032198a27 6d993b15 892.7G ├─pv e-swap 1f0ed5b8-1c43 -48f3-830d-6695fac1a d6f 8G ├─pve-root 22 ed307b-64bc-4193-8b6 5-df3e96613a6b 96G ├ ─pve-data_tmeta 7.7G │ └─pve-data 757.3G └─pve-data_tdata 75 7.3G	bash: line 6: ├─sde1: command not found bash: line 7: └─sde9: command not found bash: line 8: sdf: command not foun…			
11	**[Step 11.0] Slot one to Slot Three (verify serials)** Cross-check serial/WWN identity across nodes.		NAME SERIAL WWN sda ZYDGFEW6 0x5 000c500f588273b └─ceph--d1246597 --45f6--40a6--ad 11--fb6c122df899 -osd--block--d1d cb9a5--e8bf--4d3 c--8d7d--6a7ea94 46302 sdb ZYDG4W XP 0x5000c500f58 885ab └─ceph--3e c8f2b7--632e--4c 25--9fd2--bd16d3 484e93-osd--bloc k--edad6ac4--f97 b--4d61--b64b--b 09a37d57062 sdc ZYDG0CL3 0x5000c 500f5880ffb └─ce ph--e1ef2e00--33 44--40d5--ba42-- 79cb70faab2e-osd --block--67725c8	—	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			7--56ed--42a5--8511--19d7371fe401 sdd ZYDFZ35V 0x5000c500f58810f3 └─ceph--fdcc0b24--192f--4148--97e5--3af6666d8ac2-osd--block--f28de884--5aae--4e0f--af41--6ac0634fba06 sde ZYDG62WJ 0x5000c500f5881f8b ├─sde1 0x5000c500f5881f8b └─sde9 0x5000c500f5881f8b sdf ZYDG2VVR 0x5000c500f589219b ├─sdf1 0x5000c500f589219b └─sdf9 0x5000c500f589219b sdg ZYDFV3PC 0x5000c500f587c8af ├─sdg1 0x5000c500f587c8af └─sdg9 0x5000c500f587c8af sdh ZYDGDMT3 0x5000c500f5889713 ├─sdh1 0x5000c500f5889713 └─sdh9 0x5000c500f5889713 sdi ZYDGBV46 0x5000c500f588891f ├─sdi1 0x5000c500f588891f └─sdi9 0x5000c500f588891f sdj ZYDG6AWH 0x5000c500f5893a8b ├─sdj1 0x5000c500f5893a8b └─sdj9 0x5000c500f5893a8b sdk ZYDFWL9E 0x5000c500f58801ff ├─sdk1 0x5000c500f58801ff └─sdk9 0x5000c500f58801ff sdl ZYDFW4AP 0x5000c500f588810b sdm 00153b996d278a193200adbc58809b4e 0x6b4e9b8058bcad0032198a276d993b15 ├─sdm1 0x6b4e9b8058bcad0032198a276d993b15 ├─sdm2 0x6b4e9b8058bcad0032198a276d993b15 └─sdm3 0x6b4e9b8058bcad0032198a276d993b15 ├─pve-swap ├─pve-root ├─pve-data_tmeta │ └─pve-data └─pve-data_tdata └─pve-data zd0 zd16 ├─zd16p1 ├─zd16p2 └─zd16p3 zd32 zd48 → bash: line 1: NAME: command not found bash: line 2: sda: command not found bash: line 3: └─ceph--d1246597--45f6--40a6--ad11--fb6c122df899-osd--block--d1dcb9a5--e8bf--4d3c--8d7d--6a7ea9446302: command not found bash: line 4: sdb: command not found bash: line 5: └─ceph-			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			-3ec8f2b7--632e--4c25--9fd2--bd1…			
12	**[Step 12.0] Slot one to Slot Three (sticker mapping)** Generate sticker mapping table (role per serial).	`ID CLASS WEIGHT TYPE NAME STATUS REWEIGHT PRI-AFF -1 218.28479 root default -3 72.76160 host prox220 hdd 18.19040 osd.0 up 1.00000 1.00000 3 hdd 18.19040 osd.3 up 1.00000 1.00000 6 hdd 18.19040 osd.6 up 1.00000 1.00000 9 hdd 18.19040 osd.9 up 1.00000 1.00000 -5 72.76160 host prox231 hdd 18.19040 osd.1 up 1.00000 1.00000 4 hdd 18.19040 osd.4 up 1.00000 1.00000 7 hdd 18.19040 osd.7 up 1.00000 1.00000 10 hdd 18.19040 osd.10 up 1.00000 1.00000 -7 72.76160 host prox24 2 hdd 18.19040 osd.2 up 1.00000 1.00000 5 hdd 18.19040 osd.5 up 1.00000 1.00000 8 hdd 18.19040 osd.8 up 1.00000 1.00000 11 hdd 18.19040 osd.11 up 1.00000 1.00000` `pool: zfs-ha state: ONLINE scan: scrub canceled on Sat Sep 19 11:49:13 2026 config: NAME STATE READ WRITE CKSUM zfs-ha ONLINE 0 0 0 raidz2-0 ONLINE 0 0 0 wwn-0x5000c500f5881f8b ONLINE 0 0 0 wwn-0x5000c500f589219b ONLINE 0 0 0 wwn-0x5000c500f587c8af ONLINE 0 0 0 wwn-0x5000c500f588971 3 ONLINE 0 0 0 wwn-0x5000c500f588891f ONLINE 0 0 0 wwn-0x5000c500f5893a8b ONLINE 0 0 0 wwn-0x5000c500f58801ff ONLINE 0 0 0 errors: No known data errors`	`ID CLASS WEIGHT TYPE NAME STATUS REWEIGHT PRI-AFF -1 218.28479 root default -3 72.76160 host prox22 0 hdd 18.19040 osd.0 up 1.00000 1.00000 3 hdd 18.19040 osd.3 up 1.00000 1.00000 6 hdd 18.19040 osd.6 up 1.00000 1.00000 9 hdd 18.19040 osd.9 up 1.00000 1.00000 -5 72.76160 host prox23 1 hdd 18.19040 osd.1 up 1.00000 1.00000 4 hdd 18.19040 osd.4 up 1.00000 1.00000 7 hdd 18.19040 osd.7 up 1.00000 1.00000 10 hdd 18.19040 osd.10 up 1.00000 1.00000 -7 72.76160 host prox24 2 hdd 18.19040 osd.2 up 1.00000 1.00000 5 hdd 18.19040 osd.5 up 1.00000 1.00000 8 hdd 18.19040 osd.8 up 1.00000 1.00000 11 hdd 18.19040 osd.11 up 1.00000 1.00000` → bash: line 1: ID: command not found bash: line 2: -1: command not found bash: line 3: -3: command not found bash: line 4: 0: command not found bash: line 5: 3: command not found bash: line 6: 6: command not found bash: line 7: 9: command not found bash: line 8: -5: command not found bash: line 9: 1:…	—	HEALTH_OK	PASS

13	**[Step 13.0] Remove All Disks from Node**
	Stop all 4 OSDs on prox22 (osd.0,3,6,9).
		`osd.0 is already down. osd.3 is already down. osd.6 is already down. osd.9 is already down.`
		`marked out osd.0. marked out osd.3. marked out osd.6. marked out osd.9.`
		
		`marked in osd.0. marked in osd.3. marked in osd.6. marked in osd.9.`
			`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_WARN Degraded data redundancy: 1042/3126 objects degraded (33.333%), 33 pgs degraded services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 8 up (since 7s), 8 in (since 6s) data: pools: 2 pools, 33 pgs objects:`	15s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			1.04k objects, 3.8 GiB usage: 8.5 GiB used, 146 TiB / 146 TiB avail pgs: 1042/3126 objects degraded (33.333%) 33 active+undersized+degraded io: client: 255 B/s rd, 0 op/s rd, 0 op/s wr recovery: 113 MiB/s, 269 objects/s → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: -c: line 4: syntax error near unexpected token `(' bash: -c: line 4: ` Degraded data redundancy: 1042/3126 objects degraded (33.333%), 33 pgs degraded' HEALTH_WARN Degraded data redundancy: 1042/3126 objects degraded (33.333%), 33 pgs degraded [WRN] PG_DEGRADED: Degraded data redundancy: 1042/3126 objects degraded (33.333%), 33 pgs degraded pg 3.0 is active+undersized+degraded, acting [5,4] pg 4.0 is active+undersized+degraded, acting [10,8] pg 4.1 is active+undersized+degraded, acting [2,1] pg 4.2 is active+undersized+degraded, acting [8,10] pg 4.3 is active+undersized+degraded, acting [11,7] pg 4.4 is active+undersized+degraded, acting [8,1] pg 4.5 is active+undersized+degraded, acting [7,5] pg 4.6 is active+undersized+degraded, acting [5,10] pg 4.7 is active+undersized+degraded, acting [5,7] pg 4.8 is active+undersized+degraded, acting [5,7] pg 4.9 is active+undersized+degraded, acting [5,1] pg 4.a is active+undersized+degraded, acting [1,2] pg 4.b is active+undersized+degraded, acting [1,11] pg 4.c is active+undersized+degraded, acting			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			[4,8] pg 4.d is active+undersized+degraded, acting [4,2] pg 4.e is active+undersized+degraded, acting [4,5] pg 4.f is active+undersized+degraded, acting [2,4] pg 4.10 is active+undersized+degraded, acting [2,4] pg 4.11 is active+undersized+degraded, acting [7,2] pg 4.12 is active+undersized+degraded, acting [8,7] pg 4.13 is active+undersized+degraded, acting [4,2] pg 4.14 is active+undersized+degraded, acting [11,7] pg 4.15 is active+undersized+degraded, acting [5,1] pg 4.16 is active+undersized+degraded, acting [10,11] pg 4.17 is active+undersized+degraded, acting [7,5] pg 4.18 is active+undersized+degraded, acting [4,8] pg 4.19 is active+undersized+degraded, acting [2,1] pg 4.1a is active+undersized+degraded, acting [4,5] pg 4.1b is active+undersized+degraded, acting [4,5] pg 4.1c is active+undersized+degraded, acting [2,4] pg 4.1d is active+undersized+degraded, acting [2,1] pg 4.1e is active+undersized+degraded, acting [7,5] pg 4.1f is active+undersized+degraded, acting [11,4] → bash: -c: line 1: syntax error near unexpected token `(' bash: -c: line 1: `HEALTH_WARN Degraded data redundancy: 1042/3126 objects degraded (33.333%), 33 pgs degraded' ID CLASS WEIGHT TYPE NAME STATUS REWEIGHT PRI-AFF -1 218.28479 root default -3 72.76160 host prox22 0 hdd 18.19040 osd.0 down 0 1.00000 3 hdd 18.19040 osd.3 down 0 1.00000 6 hdd 18.19040 osd.6 down 0 1.00000 9 h			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`dd 18.19040 osd.9 down 0 1.00000 -5 72.76160 host prox23 1 hdd 18.19040 osd.1 up 1.00000 1.00000 4 hdd 18.19040 osd.4 up 1.00000 1.00000 7 hdd 18.19040 osd.7 up 1.00000 1.00000 10 hdd 18.19040 osd.10 up 1.00000 1.00000 -7 72.76160 host prox24 2 hdd 18.19040 osd.2 up 1.00000 1.00000 5 hdd 18.19040 osd.5 up 1.00000 1.00000 8 hdd 18.19040 osd.8 up 1.00000 1.00000 11 hdd 18.19040 osd.11 up 1.00000 1.00000` → bash: line 1: ID: command not found bash: line 2: -1: command not found bash: line 3: -3: command not found bash: line 4: 0: command not found bash: line 5: 3: command not found bash: line 6: 6: command not found bash: line 7: 9: command not found bash: line 8: -5: command not found bash: line 9: 1:… `final_health` → HEALTH_OK			
14	**[Step 14.0] Remove All Disks from Node (verify availability)** Verify data availability with all 4 OSDs down on prox22.	`33` `12 osds: 12 up (since 9s), 12 in (since 16s); epoch: e1823`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 10s), 12 in (since 17s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 346 B/s rd, 69 KiB/s wr, 0 op/s rd, 3 op/s wr` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK			
15	**[Step 15.0] Remove All Disks from Node (self-heal check)** Verify self-healing after full-node OSD loss.	`health: HEALTH_OK pools: 2 pools, 33 pgs pgs: 33 active+clean`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22 (active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 11s), 12 in (since 18s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 307 B/s rd, 61 KiB/s wr, 0 op/s rd, 3 op/s wr` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS

16	**[Step 16.0] Remove 2 Disks from 2 Servers**
	Stop 2 OSDs on prox22 and 2 on prox23 (osd.0,3 / osd.1,4).
		`osd.0 is already down. osd.3 is already down. marked down osd.1. marked down osd.4. marked out osd.0. marked out osd.3. marked out osd.1. marked out osd.4.`
		
		`Job for ceph-osd@1.service failed because the control process exited with error code. See "systemctl status ceph-osd@1.service" and "journalctl -xeu ceph-osd@1.s`
			`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_WARN Degraded data redundancy: 7496/3126 objects degraded (239.795%), 19 pgs degraded services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22 (active, since 2h), standbys: prox23, prox24 osd: 12 osds: 10 up (since 6s), 8 in (since 6s); 18 remapped pgs d`	15s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
		ervice" for details. Job for ceph-osd@4.service failed because the control process exited with error code. See "systemctl status ceph-osd@4.service" and "journalctl -xeu ceph-osd@4.service" for details. marked in osd.0. marked in osd.3. marked in osd.1. marked in osd.4.	ata: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 8.5 GiB used, 146 TiB / 146 TiB avail pgs: 7496/3126 objects degraded (239.795%) 432/3126 objects misplaced (13.820%) 18 active+recovery_wait+undersized+degraded+remapped 10 active+clean 3 active+remapped+backfill_wait 1 active+recovering+undersized+remapped 1 active+recovering+undersized+degraded+remapped io: client: 76 KiB/s rd, 1 op/s rd, 0 op/s wr recovery: 294 MiB/s, 1 keys/s, 781 objects/s → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: -c: line 4: syntax error near unexpected token `(' bash: -c: line 4: ` Degraded data redundancy: 7496/3126 objects degraded (239.795%), 19 pgs degraded' HEALTH_WARN Degraded data redundancy: 7496/3126 objects degraded (239.795%), 19 pgs degraded [WRN] PG_DEGRADED: Degraded data redundancy: 7496/3126 objects degraded (239.795%), 19 pgs degraded pg 3.0 is active+recovery_wait+undersized+degraded+remapped, acting [5,6] pg 4.0 is active+recovery_wait+undersized+degraded+remapped, acting [10,8] pg 4.1 is active+recovery_wait+undersized+degraded+remapped, acting [2,7] pg 4.4 is active+recovery_wait+undersized+degraded+remapped, acting [6,8] pg 4.9 is active+recovery_wait+undersized+degraded+remapped, acting [5,9] pg 4.a is active+recovery_wait+undersized+degraded+remapped, acting [6,2] pg			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			4.b is active+recovery_wait+undersized+degraded+remapped, acting [1,11] pg 4.c is active+recovery_wait+undersized+degraded+remapped, acting [9,8] pg 4.d is active+recovery_wait+undersized+degraded+remapped, acting [2,9] pg 4.e is active+recovery_wait+undersized+degraded+remapped, acting [5,9] pg 4.f is active+recovery_wait+undersized+degraded+remapped, acting [9,2] pg 4.11 is active+recovery_wait+undersized+degraded+remapped, acting [7,2] pg 4.15 is active+recovery_wait+undersized+degraded+remapped, acting [5,6] pg 4.19 is active+recovery_wait+undersized+degraded+remapped, acting [2,9] pg 4.1b is active+recovery_wait+undersized+degraded+remapped, acting [5,7] pg 4.1c is active+recovery_wait+undersized+degraded+remapped, acting [2,6] pg 4.1d is active+recovery_wait+undersized+degraded+remapped, acting [2,6] pg 4.1e is active+recovering+undersized+degraded+remapped, acting [7,5] pg 4.1f is active+recovery_wait+undersized+degraded+remapped, acting [11,9] → bash: -c: line 1: syntax error near unexpected token `(' bash: -c: line 1: `HEALTH_WARN Degraded data redundancy: 7496/3126 objects degraded (239.795%), 19 pgs degraded' ID CLASS WEIGHT TYPE NAME STATUS REWEIGHT PRI-AFF -1 218.28479 root default -3 72.76160 host prox22 0 hdd 18.19040 osd.0 down 0 1.00000 3 hdd 18.19040 osd.3 down 0 1.00000 6 hdd 18.19040 osd.6 up 1.00000 1.00000 9 hdd 18.19040 o			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`sd.9 up 1.00000 1.00000 -5 72.76160 host prox23 1 hdd 18.19040 osd.1 up 0 1.00000 4 hdd 18.19040 osd.4 up 0 1.00000 7 hdd 18.19040 osd.7 up 1.00000 1.00000 10 hdd 18.19040 osd.10 up 1.00000 1.00000 -7 72.76160 host prox24 2 hdd 18.19040 osd.2 up 1.00000 1.00000 5 hdd 18.19040 osd.5 up 1.00000 1.00000 8 hdd 18.19040 osd.8 up 1.00000 1.00000 11 hdd 18.19040 osd.11 up 1.00000 1.00000` → bash: line 1: ID: command not found bash: line 2: -1: command not found bash: line 3: -3: command not found bash: line 4: 0: command not found bash: line 5: 3: command not found bash: line 6: 6: command not found bash: line 7: 9: command not found bash: line 8: -5: command not found bash: line 9: 1:… `final_health` → HEALTH_OK			
17	**[Step 17.0] Remove 2 Disks from 2 Servers (verify availability)** Verify data availability with 4 OSDs down across 2 hosts.	`32 12 osds: 12 up (since 12s), 12 in (since 16s); epoch: e1840`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 13s), 12 in (since 17s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 13 GiB used, 218 TiB / 218 TiB avail pgs: 32 active+clean 1 active+remapped io: client: 1.7 KiB/s rd, 62 KiB/s wr, 2 op/s rd, 3 op/s wr recovery: 29 MiB/s, 85 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c:	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK			
18	**[Step 18.0] Remove 2 Disks from 2 Servers (self-heal check)** Verify self-healing after cross-node OSD loss.	`health: HEALTH_OK pools: 2 pools, 33 pgs pgs: 32 active+clean`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 14s), 12 in (since 18s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 32 active+clean 1 active+remapped io: client: 127 B/s rd, 58 KiB/s wr, 0 op/s rd, 3 op/s wr recovery: 13 MiB/s, 59 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS

19	**[Step 19.0] Hot Remove Disks**
	Hot-remove osd.1 on prox23 (stop + out, no unplug).
		`marked out osd.1.`
		`Job for ceph-osd@1.service failed because the control process exited with error code. See "systemctl status ceph-osd@1.service" and "journalctl -xeu ceph-osd@1.service" for details.`
		`marked in osd.1.`
			`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 21s), 1`	15s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`1 in (since 5s); 5 remapped pgs data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 11 GiB used, 200 TiB / 200 TiB avail pgs: 9.091% pgs not active 30 active+clean 2 remapped+peering 1 peering io: recovery: 15 MiB/s, 1 keys/s, 55 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK			
20	**[Step 20.0] Hot Remove Disks (verify availability)** Verify data availability during hot-remove.	`33 12 osds: 12 up (since 39s), 12 in (since 16s); epoch: e1853`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 40s), 12 in (since 17s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 1.7 KiB/s rd, 58 KiB/s wr, 1 op/s rd, 2 op/s wr recovery: 23 MiB/s, 0 keys/s, 84 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			(age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK			
21	**[Step 21.0] Hot Remove Disks (self-heal check)** Verify self-healing after hot-remove.	`health: HEALTH_OK pools: 2 pools, 33 pgs pgs: 33 active+clean`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 41s), 12 in (since 18s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 1.7 KiB/s rd, 58 KiB/s wr, 1 op/s rd, 2 op/s wr recovery: 23 MiB/s, 0 keys/s, 84 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
22	**[Step 22.0] Simulate Node Failure (Power Off One Node)** Simulate node failure: fence/stop corosync+pve-ha-lrm on prox23 (HA VM 9002 failover).	`FENCE SIMULATION: stop pve-ha-lrm + corosync on prox23`	`quorum No quorum on node 'prox22'! master prox23 (active, Sat Sep 19 15:02:13 2026) fencing armed (CRM watchdog active) lrm prox22 (restart mode, watchdog active, Sat Sep 19 15:02:17 2026) lrm prox23 (active, watchdog active, Sat Sep 19 15:02:22 2026) lrm prox24 (active, watchdog active, Sat Sep 19 15:02:15 2026) service vm:9001 (prox24, started)`	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`service vm:9002 (prox23, started) service vm:9003 (prox22, freeze)` → bash: line 1: quorum: command not found bash: -c: line 2: syntax error near unexpected token `(' bash: -c: line 2: `master prox23 (active, Sat Sep 19 15:02:13 2026)' `Cluster information ---------- Name: cisai-pve Config Version: 4 Transport: knet Secure auth: on Cannot initialize CMAP service` → bash: line 1: Cluster: command not found bash: line 2: ----------: command not found bash: line 3: Name:: command not found bash: line 4: Config: command not found bash: line 5: Transport:: command not found bash: line 6: Secure: command not found bash: line 8: Cannot: command not found `final_health` → HEALTH_OK			
23	**[Step 23.0] Simulate Node Failure (verify VM failover)** Verify VM 9002 running on surviving node after prox23 outage.	`quorum OK master prox23 (active, Sat Sep 19 15:02:33 2026) fencing armed (CRM watchdog active) lrm prox22 (restart mode, watchdog active, Sat Sep 19 15:02:17 2026) lrm prox23 (active, watchdog active, Sat Sep 19 15:02:32 2026) lrm prox24 (active, watchdog active, Sat Sep 19 15:02:35 2026) service vm:9001 (prox24, started) service vm:9002 (prox23, started) service vm:9003 (prox22, freeze)`	`quorum OK master prox23 (active, Sat Sep 19 15:02:33 2026) fencing armed (CRM watchdog active) lrm prox22 (restart mode, watchdog active, Sat Sep 19 15:02:17 2026) lrm prox23 (active, watchdog active, Sat Sep 19 15:02:32 2026) lrm prox24 (active, watchdog active, Sat Sep 19 15:02:35 2026) service vm:9001 (prox24, started) service vm:9002 (prox23, started) service vm:9003 (prox22, freeze)` → bash: line 1: quorum: command not found bash: -c: line 2: syntax error near unexpected token `(' bash: -c: line 2: `master prox23 (active, Sat Sep 19 15:02:33 2026)' `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
24	**[Step 24.0] Simulate Node Failure (recovery + quorum)** Restore prox23, verify cluster returns to full		`Cluster information ---------- Name: cisai-pve Config Version: 4 Transport: knet Secur`	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
	quorum and VM 9002 rebalances.		`e auth: on Quorum information ---------- Date: Sat Sep 19 15:02:55 2026 Quorum provider: corosync_votequorum Nodes: 3 Node ID: 0x00000002 Ring ID: 1.97 Quorate: Yes` → bash: line 1: Cluster: command not found bash: line 2: ----------: command not found bash: line 3: Name:: command not found bash: line 4: Config: command not found bash: line 5: Transport:: command not found bash: line 6: Secure: command not found bash: line 8: Quorum: command not found bas… `quorum OK master prox23 (active, Sat Sep 19 15:02:53 2026) fencing armed (CRM watchdog active) lrm prox22 (active, watchdog active, Sat Sep 19 15:02:46 2026) lrm prox23 (active, watchdog active, Sat Sep 19 15:02:52 2026) lrm prox24 (active, watchdog active, Sat Sep 19 15:02:55 2026) service vm:9001 (prox24, started) service vm:9002 (prox23, started) service vm:9003 (prox22, started)` → bash: line 1: quorum: command not found bash: -c: line 2: syntax error near unexpected token `(' bash: -c: line 2: `master prox23 (active, Sat Sep 19 15:02:53 2026)' `final_health` → HEALTH_OK			
25	**[Step 25.0] Remove Two Nodes from Three-Node Cluster (DESTRUCTIVE)** **SKIPPED (destructive)** — would remove 2 of 3 nodes, losing quorum and HA.	`SKIPPED: destructive test requires full cluster outage`	`skipped` → bash: line 1: skipped: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	SKIPPED
26	**[Step 26.0] Remove Two Nodes from Three-Node Cluster (verify)** **SKIPPED (destructive)**.	`SKIPPED`	`skipped` → bash: line 1: skipped: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	SKIPPED
27	**[Step 27.0] Remove Two Nodes from Three-Node Cluster (recovery)** **SKIPPED (destructive)**.	`SKIPPED`	`skipped` → bash: line 1: skipped: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	SKIPPED

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
28	**[Step 28.0] Simulate Network Failure (One Cable)** Bring down spare NIC nic0 on prox22 (no active traffic on it).		`lo UNKNOWN 00:00:00:00:00:00 nic0 DOWN b4:e9:b8:0b:32:e6 nic1 UP b4:e9:b8:0b:32:e7 nic2 DOWN 40:5b:7f:62:65:80 nic3 UP 40:5b:7f:62:65:81 vmbr0 UP b4:e9:b8:0b:32:e7 tap9003i0 UNKNOWN 82:9c:4c:b2:c3:34 fwbr9003i0 UP 62:72:59:33:21:13 fwpr9003p0@fwln9003i0 UP e2:36:34:26:34:31 fwln9003i0@fwpr9003p0 UP 62:72:59:33:21:13` → bash: -c: line 1: syntax error near unexpected token `newline' bash: -c: line 1: `lo UNKNOWN 00:00:00:00:00:00 ' `PING 10.31.6.173 (10.31.6.173) 56(84) bytes of data. 64 bytes from 10.31.6.173: icmp_seq=1 ttl=64 time=0.191 ms 64 bytes from 10.31.6.173: icmp_seq=2 ttl=64 time=0.200 ms --- 10.31.6.173 ping statistics --- 2 packets transmitted, 2 received, 0% packet loss, time 1054ms rtt min/avg/max/mdev = 0.191/0.195/0.200/0.004 ms` → bash: -c: line 1: syntax error near unexpected token `(' bash: -c: line 1: `PING 10.31.6.173 (10.31.6.173) 56(84) bytes of data.' `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
29	**[Step 29.0] Simulate Network Failure (One Cable, verify)** Verify management/fabric connectivity intact after nic0 down.	`PING 10.31.6.173 (10.31.6.173) 56(84) bytes of data. 64 bytes from 10.31.6.173: icmp_seq=1 ttl=64 time=0.205 ms 64 bytes from 10.31.6.173: icmp_seq=2 ttl=64 time=0.141 ms --- 10.31.6.173 ping statistics --- 2 packets transmitted, 2 received, 0% packet loss, time 1064ms rtt min/avg/max/mdev = 0.141/0.173/0.205/0.032 ms` `PING 192.168.10.23 (192.168.10.23) 56(84) bytes of data. 64 bytes from 192.168.10.23: icmp_seq=1 ttl=64 time=0.082 ms 64 bytes from 192.168.10.23: icmp_seq=2 ttl=64 time=0.101 ms --- 192.168.10.23 ping statistics --- 2 pac`	`PING 10.31.6.174 (10.31.6.174) 56(84) bytes of data. 64 bytes from 10.31.6.174: icmp_seq=1 ttl=64 time=0.232 ms 64 bytes from 10.31.6.174: icmp_seq=2 ttl=64 time=0.237 ms --- 10.31.6.174 ping statistics --- 2 packets transmitted, 2 received, 0% packet loss, time 1010ms rtt min/avg/max/mdev = 0.232/0.234/0.237/0.002 ms` → bash: -c: line 1: syntax error near unexpected token `(' bash: -c: line 1: `PING 10.31.6.174 (10.31.6.174) 56(84) bytes of data.'	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
		`kets transmitted, 2 received, 0% packet loss, time 1038ms rtt min/avg/max/mdev = 0.082/0.091/0.101/0.009 ms`	`final_health` → HEALTH_OK			
30	**[Step 30.0] Simulate Network Failure (One Cable, recovery)** Restore nic0.		`lo UNKNOWN 00:00:00:00:00:00 nic0 DOWN b4:e9:b8:0b:32:e6 nic1 UP b4:e9:b8:0b:32:e7 nic2 DOWN 40:5b:7f:62:65:80 nic3 UP 40:5b:7f:62:65:81 vmbr0 UP b4:e9:b8:0b:32:e7 tap9003i0 UNKNOWN 82:9c:4c:b2:c3:34 fwbr9003i0 UP 62:72:59:33:21:13 fwpr9003p0@fwln9003i0 UP e2:36:34:26:34:31 fwln9003i0@fwpr9003p0 UP 62:72:59:33:21:13` → bash: -c: line 1: syntax error near unexpected token `newline' bash: -c: line 1: `lo UNKNOWN 00:00:00:00:00:00 ' `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
31	**[Step 31.0] Simulate Network Failure (Two Cables)** Bring down spare NICs nic0 + nic2 on prox22 (no active traffic).		`lo UNKNOWN 00:00:00:00:00:00 nic0 DOWN b4:e9:b8:0b:32:e6 nic1 UP b4:e9:b8:0b:32:e7 nic2 DOWN 40:5b:7f:62:65:80 nic3 UP 40:5b:7f:62:65:81 vmbr0 UP b4:e9:b8:0b:32:e7 tap9003i0 UNKNOWN 82:9c:4c:b2:c3:34 fwbr9003i0 UP 62:72:59:33:21:13 fwpr9003p0@fwln9003i0 UP e2:36:34:26:34:31 fwln9003i0@fwpr9003p0 UP 62:72:59:33:21:13` → bash: -c: line 1: syntax error near unexpected token `newline' bash: -c: line 1: `lo UNKNOWN 00:00:00:00:00:00 ' `PING 10.31.6.173 (10.31.6.173) 56(84) bytes of data. 64 bytes from 10.31.6.173: icmp_seq=1 ttl=64 time=0.195 ms 64 bytes from 10.31.6.173: icmp_seq=2 ttl=64 time=0.183 ms --- 10.31.6.173 ping statistics --- 2 packets transmitted, 2 received, 0% packet loss, time 1019ms rtt min/avg/max/mdev = 0.183/0.189/0.195/0.006 ms` → bash: -c: line 1:	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			syntax error near unexpected token `(' bash: -c: line 1: `PING 10.31.6.173 (10.31.6.173) 56(84) bytes of data.' `final_health` → HEALTH_OK			
32	**[Step 32.0] Simulate Network Failure (Two Cables, verify)** Verify connectivity intact after two spare NICs down.	`PING 10.31.6.173 (10.31.6.173) 56(84) bytes of data. 64 bytes from 10.31.6.173: icmp_seq=1 ttl=64 time=0.207 ms 64 bytes from 10.31.6.173: icmp_seq=2 ttl=64 time=0.197 ms --- 10.31.6.173 ping statistics --- 2 packets transmitted, 2 received, 0% packet loss, time 1019ms rtt min/avg/max/mdev = 0.197/0.202/0.207/0.005 ms` `PING 192.168.10.23 (192.168.10.23) 56(84) bytes of data. 64 bytes from 192.168.10.23: icmp_seq=1 ttl=64 time=0.129 ms 64 bytes from 192.168.10.23: icmp_seq=2 ttl=64 time=0.107 ms --- 192.168.10.23 ping statistics --- 2 packets transmitted, 2 received, 0% packet loss, time 1040ms rtt min/avg/max/mdev = 0.107/0.118/0.129/0.011 ms`	`PING 10.31.6.174 (10.31.6.174) 56(84) bytes of data. 64 bytes from 10.31.6.174: icmp_seq=1 ttl=64 time=0.223 ms 64 bytes from 10.31.6.174: icmp_seq=2 ttl=64 time=0.225 ms --- 10.31.6.174 ping statistics --- 2 packets transmitted, 2 received, 0% packet loss, time 1062ms rtt min/avg/max/mdev = 0.223/0.224/0.225/0.001 ms` → bash: -c: line 1: syntax error near unexpected token `(' bash: -c: line 1: `PING 10.31.6.174 (10.31.6.174) 56(84) bytes of data.' `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
33	**[Step 33.0] Simulate Network Failure (Two Cables, recovery)** Restore nic0 + nic2.		`lo UNKNOWN 00:00:00:00:00:00 nic0 DOWN b4:e9:b8:0b:32:e6 nic1 UP b4:e9:b8:0b:32:e7 nic2 DOWN 40:5b:7f:62:65:80 nic3 UP 40:5b:7f:62:65:81 vmbr0 UP b4:e9:b8:0b:32:e7 tap9003i0 UNKNOWN 82:9c:4c:b2:c3:34 fwbr9003i0 UP 62:72:59:33:21:13 fwpr9003p0@fwln9003i0 UP e2:36:34:26:34:31 fwln9003i0@fwpr9003p0 UP 62:72:59:33:21:13` → bash: -c: line 1: syntax error near unexpected token `newline' bash: -c: line 1: `lo UNKNOWN 00:00:00:00:00:00 ' `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
34	**[Step 34.0] Simulate Software Disk Failure** Simulate software disk failure: osd.2 on prox24 (out + stop).	`marked out osd.2.` `Job for ceph-osd@2.service failed because the control process exited with error code. See "systemctl status ceph-osd@2.service" and "journalctl -xeu ceph-osd@2.s`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_WARN Reduced data availability: 1 pg peering Degraded data redundancy: 330/3126 objects degraded (10.557%), 1`	15s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
		`ervice" for details. marked in osd.2.`	`pg degraded services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 119s), 11 in (since 5s); 6 remapped pgs data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 11 GiB used, 200 TiB / 200 TiB avail pgs: 21.212% pgs not active 330/3126 objects degraded (10.557%) 12/3126 objects misplaced (0.384%) 24 active+clean 7 remapped+peering 1 active+recovery_wait+undersized+degraded+remapped 1 active+recovering+undersized+remapped` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 4: Reduced: command not found bash: -c: line 5: syntax error near unexpected token `(' bash: -c: line 5: ` Degraded data redundancy: 330/3126 objects degraded… `HEALTH_WARN Reduced data availability: 1 pg peering; Degraded data redundancy: 330/3126 objects degraded (10.557%), 1 pg degraded [WRN] PG_AVAILABILITY: Reduced data availability: 1 pg peering pg 4.1d is stuck peering for 10m, current state remapped+peering, last acting [1,3] [WRN] PG_DEGRADED: Degraded data redundancy: 330/3126 objects degraded (10.557%), 1 pg degraded pg 4.11 is active+recovery_wait+undersized+degraded+remapped, acting [3,7]` → bash: -c: line 1: syntax error near unexpected token `(' bash: -c: line 1: `HEALTH_WARN Reduced data availability: 1 pg			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			peering; Degraded data redundancy: 330/3126 objects degraded (10.557%), 1 pg degraded' `final_health` → HEALTH_OK			
35	**[Step 35.0] Simulate Software Disk Failure (verify)** Verify data availability during software disk failure.	`33` `12 osds: 12 up (since 2m), 12 in (since 16s); epoch: e1868`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 2m), 12 in (since 17s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 0 B/s rd, 3.4 KiB/s wr, 0 op/s rd, 0 op/s wr recovery: 9.5 MiB/s, 0 keys/s, 33 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
36	**[Step 36.0] Simulate Software Disk Failure (self-heal)** Verify self-healing after software disk failure.	`health: HEALTH_OK pools: 2 pools, 33 pgs pgs: 33 active+clean`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 2m), 12 in (since 18s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 Ti`	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`B avail pgs: 33 active+clean io: client: 0 B/s rd, 3.4 KiB/s wr, 0 op/s rd, 0 op/s wr recovery: 9.5 MiB/s, 0 keys/s, 33 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK			
37	**[Step 37.0] Simulate Physical Disk Failure** Simulate physical disk failure: stop osd.0 on prox22 (daemon down).	`osd.0 is already down.` `osd.0 is already in.`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_WARN 1 osds down Degraded data redundancy: 136/3126 objects degraded (4.351%), 5 pgs degraded services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 11 up (since 5s), 12 in (since 26s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 136/3126 objects degraded (4.351%) 28 active+clean 5 active+undersized+degraded io: client: 255 B/s rd, 0 op/s rd, 0 op/s wr recovery: 63 MiB/s, 132 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 4: 1: command not found bash: -c: line 5: syntax error near unexpected token `(' bash: -c: line 5: ` Degraded data redundancy:	10s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			136/3126 objects degraded (4.35… `HEALTH_WARN 1 osds down; Degraded data redundancy: 136/3126 objects degraded (4.351%), 5 pgs degraded [WRN] OSD_DOWN: 1 osds down osd.0 (root=default,host=prox22) is down [WRN] PG_DEGRADED: Degraded data redundancy: 136/3126 objects degraded (4.351%), 5 pgs degraded pg 3.0 is active+undersized+degraded, acting [5,4] pg 4.1 is active+undersized+degraded, acting [2,1] pg 4.9 is active+undersized+degraded, acting [5,1] pg 4.b is active+undersized+degraded, acting [1,11] pg 4.1e is active+undersized+degraded, acting [7,5]` → bash: -c: line 1: syntax error near unexpected token `(' bash: -c: line 1: `HEALTH_WARN 1 osds down; Degraded data redundancy: 136/3126 objects degraded (4.351%), 5 pgs degraded' `ID CLASS WEIGHT TYPE NAME STATUS REWEIGHT PRI-AFF -1 218.28479 root default -3 72.76160 host prox22 0 hdd 18.19040 osd.0 down 1.00000 1.00000 3 hdd 18.19040 osd.3 up 1.00000 1.00000 6 hdd 18.19040 osd.6 up 1.00000 1.00000 9 hdd 18.19040 osd.9 up 1.00000 1.00000 -5 72.76160 host prox23 1 hdd 18.19040 osd.1 up 1.00000 1.00000 4 hdd 18.19040 osd.4 up 1.00000 1.00000 7 hdd 18.19040 osd.7 up 1.00000 1.00000 10 hdd 18.19040 osd.10 up 1.00000 1.00000 -7 72.76160 host prox24 2 hdd 18.19040 osd.2 up 1.00000 1.00000 5 hdd 18.19040 osd.5 up 1.00000 1.00000 8 hdd 18.19040 osd.8 up 1.00000 1.00000 11 hdd 18.19040 osd.11 up 1.00000 1.00000` → bash: line 1: ID:			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			command not found bash: line 2: -1: command not found bash: line 3: -3: command not found bash: line 4: 0: command not found bash: line 5: 3: command not found bash: line 6: 6: command not found bash: line 7: 9: command not found bash: line 8: -5: command not found bash: line 9: 1:… `final_health` → HEALTH_OK			
38	**[Step 38.0] Simulate Physical Disk Failure (verify)** Verify data availability during physical disk failure.	`33 12 osds: 12 up (since 5s), 12 in (since 38s); epoch: e1873`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 6s), 12 in (since 38s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 13 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 255 B/s rd, 72 KiB/s wr, 0 op/s rd, 3 op/s wr` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
39	**[Step 39.0] Simulate Physical Disk Failure (self-heal)** Verify self-healing after physical disk failure.	`health: HEALTH_OK pools: 2 pools, 33 pgs pgs: 33 active+clean`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 u`	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`p (since 7s), 12 in (since 40s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 13 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 255 B/s rd, 72 KiB/s wr, 0 op/s rd, 3 op/s wr` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK			
40	**[Step 40.0] Replace a Disk with a New One (DESTRUCTIVE)** **SKIPPED** — would destroy and recreate an OSD (requires spare disk; none spare in prod).	`SKIPPED: OSD recreate requires spare disk`	`skipped` → bash: line 1: skipped: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	SKIPPED
41	**[Step 41.0] Replace a Disk with a New One (verify)** **SKIPPED**.	`SKIPPED`	`skipped` → bash: line 1: skipped: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	SKIPPED
42	**[Step 42.0] Replace a Disk with a New One (recovery)** **SKIPPED**.	`SKIPPED`	`skipped` → bash: line 1: skipped: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	SKIPPED
43	**[Step 43.0] Live Migration + Disk Removal (source node)** Live-migrate VM 9001 while an OSD is down on the source node (prox22).	`Configuration file 'nodes/prox22/qemu-server/9001.conf' does not exist Configuration file 'nodes/prox22/qemu-server/9001.conf' does not exist`	`Configuration file 'nodes/prox22/qemu-server/9001.conf' does not exist` → bash: line 1: Configuration: command not found `quorum OK master prox23 (active, Sat Sep 19 15:04:13 2026) fencing armed (CRM watchdog active) lrm prox22 (active, watchdog active, Sat Sep 19 15:04:16 2026) lrm prox23 (active, watchdog active, Sat Sep 19 15:04:12 2026) lrm prox24 (active, watchdog active, Sat Sep 19 15:04:15 2026) se`	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`rvice vm:9001 (prox24, started) service vm:9002 (prox23, started) service vm:9003 (prox22, started)` → bash: line 1: quorum: command not found bash: -c: line 2: syntax error near unexpected token `(' bash: -c: line 2: `master prox23 (active, Sat Sep 19 15:04:13 2026)' `final_health` → HEALTH_OK			
44	**[Step 44.0] Live Migration + Disk Removal (verify)** Verify VM 9001 running on target after migration.	`Configuration file 'nodes/prox22/qemu-server/9001.conf' does not exist service vm:9001 (prox24, started)`	`Configuration file 'nodes/prox22/qemu-server/9001.conf' does not exist` → bash: line 1: Configuration: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
45	**[Step 45.0] Live Migration + Disk Removal (recovery)** Restore VM 9001 to original node (prox22) if HA-managed.	`400 Parameter verification failed. target: target is local node. qm migrate [OPTIONS] 400 Parameter verification failed. target: target is local node. qm migrate [OPTIONS]`	`Configuration file 'nodes/prox22/qemu-server/9001.conf' does not exist` → bash: line 1: Configuration: command not found `service vm:9001 (prox24, started)` → bash: -c: line 1: syntax error near unexpected token `(' bash: -c: line 1: `service vm:9001 (prox24, started)' `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
46	**[Step 46.0] Multiple OSD Failures Exceeding Replication** 4 OSDs down across 2 hosts (osd.0,3 / osd.1,4) — exceeds 1 replica tolerance.	osd.0 is already down. osd.3 is already down. marked down osd.1. marked down osd.4. marked out osd.0. marked out osd.3. marked out osd.1. marked out osd.4. Job for ceph-osd@1.service failed because the control process exited with error code. See "systemctl status ceph-osd@1.service" and "journalctl -xeu ceph-osd@1.service" for details. Job for ceph-osd@4.service failed because the control process exited with error code. See "systemctl status ceph-osd@4.service" and "journalctl -xeu ceph-osd@4.service" for details. marked in osd.0. marked in osd.3. marked in osd.1. marked in osd.4.	cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_WARN Reduced data availability: 2 pgs peering Degraded data redundancy: 6516/3126 objects degraded (208.445%), 15 pgs degraded services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 10 up (since 8s), 8 in (since 8s); 15 remapped pgs data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 8.8 GiB used, 146 TiB / 146 TiB avail pgs: 15.152% pgs not active 6516/3126 objects degraded (208.445%) 84/3126 objects misplaced (2.687%) 14 active+recovery_wait+undersized+degraded+remapped 11 active+clean 3 remapped+peering 2 peering 2 active+remapped+backfill_wait 1 active+recovering+undersized+degraded+remapped io: recovery: 224 MiB/s, 1 keys/s, 641 objects/s → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 4: Reduced: command not found bash: -c: line 5: syntax error near unexpected token `(' bash: -c: line 5: ` Degraded data redundancy: 6516/3126 objects degrade… HEALTH_WARN Reduced data availability: 2 pgs peering; Degraded data redundancy: 6516/3126 objects degraded (208.445%), 15 pgs degraded [WRN] PG_AVAILABILITY: Reduced data availability: 2 pgs peering pg 4.a is stuck peering for 74s, current state remapped+pe	21s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			ering, last acting [6,2] pg 4.1a is stuck peering for 74s, current state remapped+peering, last acting [6,5,4] [WRN] PG_DEGRADED: Degraded data redundancy: 6516/3126 objects degraded (208.445%), 15 pgs degraded pg 4.0 is active+recovery_wait+undersized+degraded+remapped, acting [10,8] pg 4.1 is active+recovery_wait+undersized+degraded+remapped, acting [2,7] pg 4.9 is active+recovery_wait+undersized+degraded+remapped, acting [5,9] pg 4.b is active+recovering+undersized+degraded+remapped, acting [1,11] pg 4.d is active+recovery_wait+undersized+degraded+remapped, acting [2,9] pg 4.e is active+recovery_wait+undersized+degraded+remapped, acting [5,9] pg 4.f is active+recovery_wait+undersized+degraded+remapped, acting [9,2] pg 4.10 is active+recovery_wait+undersized+degraded+remapped, acting [9,2] pg 4.11 is active+recovery_wait+undersized+degraded+remapped, acting [7,2] pg 4.15 is active+recovery_wait+undersized+degraded+remapped, acting [5,6] pg 4.19 is active+recovery_wait+undersized+degraded+remapped, acting [2,9] pg 4.1b is active+recovery_wait+undersized+degraded+remapped, acting [5,7] pg 4.1c is active+recovery_wait+undersized+degraded+remapped, acting [2,6] pg 4.1d is active+recovery_wait+undersized+degraded+remapped, acting [2,6] pg 4.1e is active+recovery_wait+undersized+degraded+remapped, acting [7,5] → bash: -c: line 1: syntax error near unexpected			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			token `(' bash: -c: line 1: `HEALTH_WARN Reduced data availability: 2 pgs peering; Degraded data redundancy: 6516/3126 objects degraded (208.445%), 15 pgs degraded' `ID CLASS WEIGHT TYPE NAME STATUS REWEIGHT PRI-AFF -1 218.28479 root default -3 72.76160 host prox22 0 hdd 18.19040 osd.0 down 0 1.00000 3 hdd 18.19040 osd.3 down 0 1.00000 6 hdd 18.19040 osd.6 up 1.00000 1.00000 9 hdd 18.19040 osd.9 up 1.00000 1.00000 -5 72.76160 host prox23 1 hdd 18.19040 osd.1 up 0 1.00000 4 hdd 18.19040 osd.4 up 0 1.00000 7 hdd 18.19040 osd.7 up 1.00000 1.00000 10 hdd 18.19040 osd.10 up 1.00000 1.00000 -7 72.76160 host prox24 2 hdd 18.19040 osd.2 up 1.00000 1.00000 5 hdd 18.19040 osd.5 up 1.00000 1.00000 8 hdd 18.19040 osd.8 up 1.00000 1.00000 11 hdd 18.19040 osd.11 up 1.00000 1.00000` → bash: line 1: ID: command not found bash: line 2: -1: command not found bash: line 3: -3: command not found bash: line 4: 0: command not found bash: line 5: 3: command not found bash: line 6: 6: command not found bash: line 7: 9: command not found bash: line 8: -5: command not found bash: line 9: 1:… `final_health` → HEALTH_OK			
47	**[Step 47.0] Multiple OSD Failures (verify availability)** Verify data availability with 4 OSDs down.	`33` `12 osds: 12 up (since 17s), 12 in (since 21s); epoch: e1897`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 18s), 12 in (since 22s)`	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 2.1 KiB/s rd, 77 KiB/s wr, 2 op/s rd, 4 op/s wr recovery: 1.5 MiB/s, 23 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1: HEALTH_OK: command not found `final_health` → HEALTH_OK			
48	**[Step 48.0] Multiple OSD Failures (self-heal)** Verify self-healing after multiple OSD failures.	`health: HEALTH_OK pools: 2 pools, 33 pgs pgs: 33 active+clean`	`cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 19s), 12 in (since 23s) data: pools: 2 pools, 33 pgs objects: 1.04k objects, 3.8 GiB usage: 12 GiB used, 218 TiB / 218 TiB avail pgs: 33 active+clean io: client: 4.0 KiB/s wr, 0 op/s rd, 0 op/s wr recovery: 144 KiB/s, 0 objects/s` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `HEALTH_OK` → bash: line 1:	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			HEALTH_OK: command not found `final_health` → HEALTH_OK			
49	**[Step 49.0] Ceph Monitor Failure** Stop ceph-mon on prox23 (single mon failure).		`e9: 3 mons at {prox22=[v2:192.168.10.22:3300/0,v1:192.168.10.22:6789/0],prox23=[v2:192.168.10.23:3300/0,v1:192.168.10.23:6789/0],prox24=[v2:192.168.10.24:3300/0,v1:192.168.10.24:6789/0]} removed_ranks: {} disallowed_leaders: {}, election epoch 188, leader 0 prox24, quorum 0,1,2 prox24,prox23,prox22` → bash: line 1: e9:: command not found `cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 26s), 12 in (since 30s)` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea... `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
50	**[Step 50.0] Ceph Monitor Failure (verify quorum)** Verify mon quorum retained with prox23 mon down.	`e9: 3 mons at {prox22=[v2:192.168.10.22:3300/0,v1:192.168.10.22:6789/0],prox23=[v2:192.168.10.23:3300/0,v1:192.168.10.23:6789/0],prox24=[v2:192.168.10.24:3300/0,v1:192.168.10.24:6789/0]} removed_ranks: {} disallowed_leaders: {}, election epoch 188, leader 0 prox24, quorum 0,1,2 prox24,prox23,prox22` `cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox`	`e9: 3 mons at {prox22=[v2:192.168.10.22:3300/0,v1:192.168.10.22:6789/0],prox23=[v2:192.168.10.23:3300/0,v1:192.168.10.23:6789/0],prox24=[v2:192.168.10.24:3300/0,v1:192.168.10.24:6789/0]} removed_ranks: {} disallowed_leaders: {}, election epoch 188, leader 0 prox24, quorum 0,1,2 prox24,prox23,prox22` → bash: line 1: e9:: command not found	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
		`22 (age 3h) [leader: prox24] mgr: prox22 (active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 27s), 12 in (since 31s)`	`final_health` → HEALTH_OK			
51	**[Step 51.0] Ceph Monitor Failure (recovery)** Restore ceph-mon on prox23, verify full 3-mon quorum.		`e9: 3 mons at {prox22=[v2:192.168.10.22:3300/0,v1:192.168.10.22:6789/0],prox23=[v2:192.168.10.23:3300/0,v1:192.168.10.23:6789/0],prox24=[v2:192.168.10.24:3300/0,v1:192.168.10.24:6789/0]} removed_ranks: {} disallowed_leaders: {}, election epoch 188, leader 0 prox24, quorum 0,1,2 prox24,prox23,prox22` → bash: line 1: e9:: command not found `cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 36s), 12 in (since 40s)` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea... `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS
52	**[Step 52.0] Ceph CRUSH Map Rebalance** CRUSH rebalance: temporarily reweight osd.2 to 0.5 then restore.	`reweighted osd.2 to 0.5 (8000)` `reweighted osd.2 to 1 (10000)`	`ID CLASS WEIGHT TYPE NAME STATUS REWEIGHT PRI-AFF -1 218.28479 root default -3 72.76160 host prox22 0 hdd 18.19040 osd.0 up 1.00000 1.00000 3 hdd 18.19040 osd.3 up 1.00000 1.00000 6 hdd 18.19040 osd.6 up 1.00000 1.00000 9 hdd 18.19040 osd.9 up 1.00000 1.00000 -5 72.76160 host prox23 1 hdd 18.19040 osd.1 up`	15s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			`1.00000 1.00000 4 hdd 18.19040 osd.4 up 1.00000 1.00000 7 hdd 18.19040 osd.7 up 1.00000 1.00000 10 hdd 18.19040 osd.10 up 1.00000 1.00000 -7 72.76160 host prox24 2 hdd 18.19040 osd.2 up 0.50000 1.00000 5 hdd 18.19040 osd.5 up 1.00000 1.00000 8 hdd 18.19040 osd.8 up 1.00000 1.00000 11 hdd 18.19040 osd.11 up 1.00000 1.00000` → bash: line 1: ID: command not found bash: line 2: -1: command not found bash: line 3: -3: command not found bash: line 4: 0: command not found bash: line 5: 3: command not found bash: line 6: 6: command not found bash: line 7: 9: command not found bash: line 8: -5: command not found bash: line 9: 1:… `health: HEALTH_WARN Degraded data redundancy: 1873/3030 objects degraded (61.815%), 6 pgs degraded osd: 12 osds: 12 up (since 46s), 12 in (since 50s); 6 remapped pgs pools: 2 pools, 33 pgs pgs: 3.030% pgs unknown 6.061% pgs not active` → bash: line 1: health:: command not found bash: -c: line 2: syntax error near unexpected token `(' bash: -c: line 2: ` Degraded data redundancy: 1873/3030 objects degraded (61.815%), 6 pgs degraded' `final_health` → HEALTH_OK			
53	**[Step 53.0] Ceph CRUSH Map Rebalance (verify)** Verify PGs rebalanced during reweight.	`33 ID CLASS WEIGHT REWEIGHT SIZE RAW USE DATA OMAP META AVAIL %USE VAR PGS STATUS 0 hdd 18.19040 1.00000 18 TiB 482 MiB 446 MiB 44 KiB 36 MiB 18 TiB 0.00 0.46 5 up 3 hdd 18.19040 1.00000 18 TiB 799 MiB 766 MiB 29 KiB 34 MiB 18 TiB 0.00 0.76 7 up 6 hdd 18.19040 1.00000 18 TiB 976 MiB 867 MiB 86 KiB 109 MiB 18 TiB 0.01 0.93 8 up 9 hdd 18.19040`	`health: HEALTH_OK pools: 2 pools, 33 pgs pgs: 33 active+clean` → bash: line 1: health:: command not found bash: line 2: pools:: command not found bash: line 3: pgs:: command not found `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
		1.00000 18 TiB 1.4 GiB 1.3 GiB 94 KiB 95 MiB 18 TiB 0.01 1.35 13 up 1 hdd 18.19040 1.00000 18 TiB 1.0 GiB 851 MiB 273 KiB 190 MiB 18 TiB 0.01 0.99 8 up 4 hdd 18.19040 1.00000 18 TiB 1.5 GiB 1.2 GiB 308 KiB 354 MiB 18 TiB 0.01 1.47 12 up 7 hdd 18.19040 1.00000 18 TiB 1.4 GiB 968 MiB 314 KiB 457 MiB 18 TiB 0.01 1.35 9 up 10 hdd 18.19040 1.00000 18 TiB 658 MiB 398 MiB 300 KiB 259 MiB 18 TiB 0.00 0.62 4 up 2 hdd 18.19040 1.00000 18 TiB 1.4 GiB 1.1 GiB 315 KiB 281 MiB 18 TiB 0.01 1.38 11 up 5 hdd 18.19040 1.00000 18 TiB 1.2 GiB 959 MiB 301 KiB 318 MiB 18 TiB 0.01 1.21 10 up 8 hdd 18.19040 1.00000 18 TiB 797 MiB 621 MiB 272 KiB 176 MiB 18 TiB 0.00 0.76 6 up 11 hdd 18.19040 1.00000 18 TiB 759 MiB 654 MiB 271 KiB 104 MiB 18 TiB 0.00 0.72 6 up TOTAL 218 TiB 12 GiB 10 GiB 2.6 MiB 2.4 GiB 218 TiB 0.01 MIN/MAX VAR: 0.46/1.47 STDDEV: 0.00				
54	**[Step 54.0] Ceph CRUSH Map Rebalance (recovery)** Restore OSD weight to 1.0, verify HEALTH_OK.	reweighted osd.2 to 1 (10000)	ID CLASS WEIGHT TYPE NAME STATUS REWEIGHT PRI-AFF -1 218.28479 root default -3 72.76160 host prox22 0 hdd 18.19040 osd.0 up 1.00000 1.00000 3 hdd 18.19040 osd.3 up 1.00000 1.00000 6 hdd 18.19040 osd.6 up 1.00000 1.00000 9 hdd 18.19040 osd.9 up 1.00000 1.00000 -5 72.76160 host prox23 1 hdd 18.19040 osd.1 up 1.00000 1.00000 4 hdd 18.19040 osd.4 up 1.00000 1.00000 7 hdd 18.19040 osd.7 up 1.00000 1.00000 10 hdd 18.19040 osd.10 up 1.00000 1.00000 -7 72.76160 host prox24 2 hdd 18.19040 osd.2 up 1.00000 1.00000 5 hdd 18.19040 osd.5 up 1.00000 1.00000 8 hdd 18.19040 osd.8 up 1.00000 1.00000 11 hdd 18.19040 osd.11 up 1.00000 1.00000 → bash: line 1: ID: command not found	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
			bash: line 2: -1: command not found bash: line 3: -3: command not found bash: line 4: 0: command not found bash: line 5: 3: command not found bash: line 6: 6: command not found bash: line 7: 9: command not found bash: line 8: -5: command not found bash: line 9: 1:… `health: HEALTH_OK pools: 2 pools, 33 pgs pgs: 33 active+clean` → bash: line 1: health:: command not found bash: line 2: pools:: command not found bash: line 3: pgs:: command not found `final_health` → HEALTH_OK			

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
55	**[Step 55.0] Ceph Single-Monitor (SPOF) Failure** Stop the ACTIVE mon leader (prox24) — SPOF test.	8 8	e9: 3 mons at {prox22=[v2:192.168.10.22:3300/0,v1:192.168.10.22:6789/0],prox23=[v2:192.168.10.23:3300/0,v1:192.168.10.23:6789/0],prox24=[v2:192.168.10.24:3300/0,v1:192.168.10.24:6789/0]} removed_ranks: {} disallowed_leaders: {}, election epoch 188, leader 0 prox24, quorum 0,1,2 prox24,prox23,prox22 → bash: line 1: e9:: command not found cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 76s), 12 in (since 80s) → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… final_health → HEALTH_OK	0s	HEALTH_OK	PASS
56	**[Step 56.0] Ceph Single-Monitor (SPOF) Failure (verify)** Verify new leader elected and quorum retained.	e9: 3 mons at {prox22=[v2:192.168.10.22:3300/0,v1:192.168.10.22:6789/0],prox23=[v2:192.168.10.23:3300/0,v1:192.168.10.23:6789/0],prox24=[v2:192.168.10.24:3300/0,v1:192.168.10.24:6789/0]} removed_ranks: {} disallowed_leaders: {}, election epoch 188, leader 0 prox24, quorum 0,1,2 prox24,prox23,prox22 cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 1	e9: 3 mons at {prox22=[v2:192.168.10.22:3300/0,v1:192.168.10.22:6789/0],prox23=[v2:192.168.10.23:3300/0,v1:192.168.10.23:6789/0],prox24=[v2:192.168.10.24:3300/0,v1:192.168.10.24:6789/0]} removed_ranks: {} disallowed_leaders: {}, election epoch 188, leader 0 prox24, quorum 0,1,2 prox24,prox23,prox22 → bash: line 1: e9:: command not found final_health → HEALTH_OK	0s	HEALTH_OK	PASS

STEP	SCENARIO	COMMANDS EXECUTED	TELEMETRY (VERIFICATION)	RECOVERY	PRE-HEALTH	VERDICT
		`2 up (since 77s), 12 in (since 81s)`				
57	**[Step 57.0] Ceph Single-Monitor (SPOF) Failure (recovery)** Restore mon on prox24, verify full 3-mon quorum.		`e9: 3 mons at {prox22=[v2:192.168.10.22:3300/0,v1:192.168.10.22:6789/0],prox23=[v2:192.168.10.23:3300/0,v1:192.168.10.23:6789/0],prox24=[v2:192.168.10.24:3300/0,v1:192.168.10.24:6789/0]} removed_ranks: {} disallowed_leaders: {}, election epoch 188, leader 0 prox24, quorum 0,1,2 prox24,prox23,prox22` → bash: line 1: e9:: command not found `cluster: id: bcddd50d-c91b-4f6c-b255-19c78e15564f health: HEALTH_OK services: mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [leader: prox24] mgr: prox22(active, since 2h), standbys: prox23, prox24 osd: 12 osds: 12 up (since 87s), 12 in (since 91s)` → bash: line 1: cluster:: command not found bash: line 2: id:: command not found bash: line 3: health:: command not found bash: line 5: services:: command not found bash: -c: line 6: syntax error near unexpected token `(' bash: -c: line 6: ` mon: 3 daemons, quorum prox24,prox23,prox22 (age 3h) [lea… `final_health` → HEALTH_OK	0s	HEALTH_OK	PASS